

Generating M/F Slopes From Regression

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Intro

I've managed to resolve the puzzle I was struggling with. The first part of this describes how to use the model coefficients to generate the slopes “by gender”, depending on whether Gender is coded as a dummy variable (0/1) or as a Deviation coded variable (-1/1). I then illustrate how this leads to identical results for the *unstandardised* coefficients (which is what I expected). Then I demonstrate how it seems to be broken for the *standardised* coefficients.

Then I turn to the mistake I've made: *I forgot that Gender is also standardized* in the standardized models.

The Model

The basic model for our regressions looks like this (ignoring control variables for simplicity):

$$Reading = \beta_0 + \beta_G * Gender + \beta_E * EH + \beta_I * Gender * EH + \epsilon$$

where β_0 is an intercept, β_G is the Gender coefficient, β_E is the EH coefficient, and β_I is the interaction coefficient.

Since we're interested in the EH slope overall, we can rewrite that as:

$$Reading = \beta_0 + \beta_G * Gender + (\beta_E + \beta_I * Gender) * EH + \epsilon$$

This means that for each gender, the slope for EH is given by:

$$(\beta_E + \beta_I * Gender)$$

The coding schemes

Depending on which coding scheme we use, the formula will look a little different:

Dummy (or Treatment) Coding (0/1) If we dummy code the variable (as done originally by default), then we have the following:

$$\begin{aligned} \text{Girls: } Slope &= (\beta_E + \beta_I * Gender) = (\beta_E + \beta_I * 0) = \beta_E \\ \text{Boys: } Slope &= (\beta_E + \beta_I * Gender) = (\beta_E + \beta_I * 1) = \beta_E + \beta_I \end{aligned}$$

Deviation (or Sum) Coding (-1/1) If we deviation code the variable, then we have the following instead:

$$\text{Girls: } Slope = (\beta_E + \beta_I * Gender) = (\beta_E + \beta_I * -1) = \beta_E - \beta_I$$

$$\text{Boys: } Slope = (\beta_E + \beta_I * Gender) = (\beta_E + \beta_I * 1) = \beta_E + \beta_I$$

Comparing the estimates If we consider the results for Adaptability (Unstandardized coefficients), everything works out nicely:

Under the Dummy Coding scheme:

$$\text{Girls: } = \beta_E = .302; \text{ Boys: } = \beta_E + \beta_I = .302 - .455 = -.153$$

Under the Deviation Coding:

$$\text{Girls: } = \beta_E - \beta_I = .076 - (-.229) = .305; \text{ Boys: } = \beta_E + \beta_I = .076 + (-.229) = -.153$$

(a small difference in girls is likely just because of the rounding of the coefficients)

Why I was Puzzled:

What was puzzling to me is that the Standardized coefficients aren't always behaving the same way (staying with Adaptability):

Under the Dummy Coding scheme:

$$\text{Girls: } = \beta_E = .173; \text{ Boys: } = \beta_E + \beta_I = .173 - .201 = -.028$$

Under the Deviation Coding:

$$\text{Girls: } = \beta_E - \beta_I = .043 - (-.131) = .174; \text{ Boys: } = \beta_E + \beta_I = .043 + (-.131) = -.088$$

The difference in the boys' coefficients (-.028 vs -.088) can't just be down to rounding.

The answer: In the above, I have forgotten that the standardized analysis is not using Gender as -1/1 or 0/1, but rather a standardized version of the Gender values. If the data were balanced between genders, then the scaled version would be relatively close to -1/1, but in our case there is an imbalance. Consequently the formulas I generated for the slopes don't work for standardised coefficients where the scaling might produce something like "Girls = -1.3, Boys = .7".

Final Word

The standardized coefficients, using -1/1 are correct and should be what we use for the paper.

Phew.